In [1]:
```python
import pandas as pd
df = pd.read_csv("https://raw.githubusercontent.com/blacktreeM/econ/refs/heads/main
```

In [3]:
```python
df.describe()
```

Out[3]:

	G	W	ADJOE	ADJDE	BARTHAG	EFG_O	
count	4249.000000	4249.000000	4249.000000	4249.000000	4249.000000	4249.000000	4249
mean	30.753589	16.177924	103.625865	103.627418	0.493439	50.006354	50
std	3.758512	6.497419	7.402858	6.563856	0.255445	3.080054	2
min	5.000000	0.000000	76.600000	84.000000	0.005000	39.200000	39
25%	29.000000	11.000000	98.600000	99.000000	0.281200	48.000000	48
50%	31.000000	16.000000	103.300000	103.800000	0.472500	50.000000	50
75%	33.000000	21.000000	108.300000	108.300000	0.715300	52.000000	52
max	40.000000	38.000000	130.600000	125.300000	0.984200	61.000000	60

8 rows × 21 columns

In [4]:
```python
print(df.columns.tolist())
# ['TEAM', 'CONF', 'G', 'W', 'ADJOE', 'ADJDE', 'BARTHAG', 'EFG_O', 'EFG_D', 'TOR',
```

```
['TEAM', 'CONF', 'G', 'W', 'ADJOE', 'ADJDE', 'BARTHAG', 'EFG_O', 'EFG_D', 'TOR', 'TO
RD', 'ORB', 'DRB', 'FTR', 'FTRD', '2P_O', '2P_D', '3P_O', '3P_D', 'ADJ_T', 'WAB', 'P
OSTSEASON', 'SEED', 'YEAR']
```

['TEAM', 'CONF', 'G', 'W', 'ADJOE', 'ADJDE', 'BARTHAG', 'EFG_O', 'EFG_D', 'TOR', 'TORD', 'ORB', 'DRB', 'FTR', 'FTRD', '2P_O', '2P_D', '3P_O', '3P_D', 'ADJ_T', 'WAB', 'POSTSEASON', 'SEED', 'YEAR'] Effective Field Goal % (EFG_O) Turnover % (TOR) Offensive Rebound % (ORB) Free Throw Rate (FTR)

She specifically wants to compare the slopes of the regression lines to see how these relationships change over time.

Task: Create similar visualizations in the next four slides

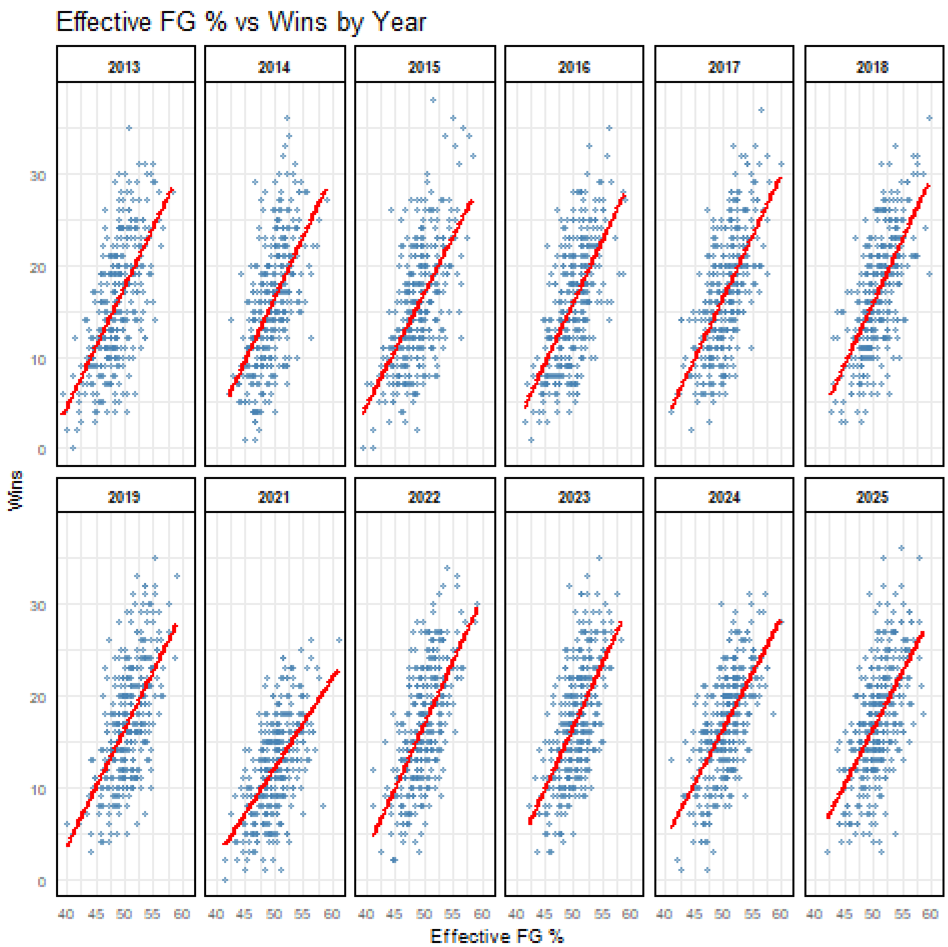
In [5]:
```python
# run this by itself to run r
%load_ext rpy2.ipython
```

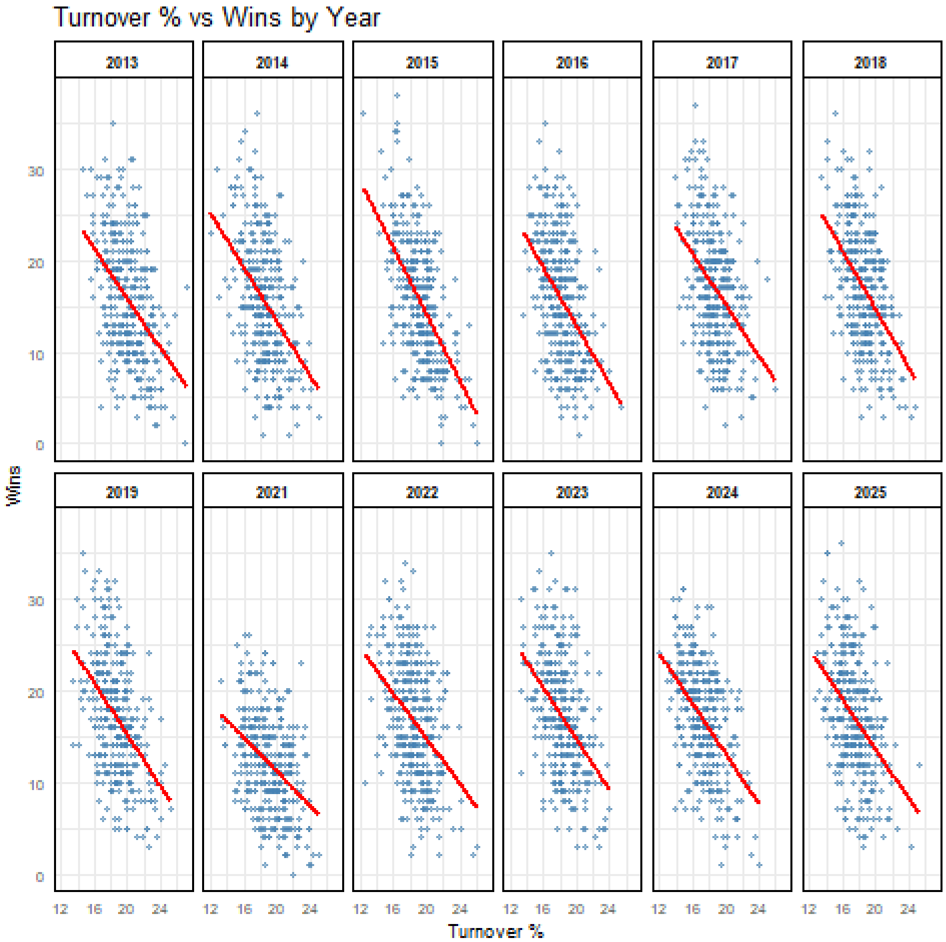
```
Error importing in API mode: ImportError('On Windows, cffi mode "ANY" is only "AB
I".')
Trying to import in ABI mode.
```

In [6]:
```r
%%R -i df
library(dplyr)
library(ggplot2)
```

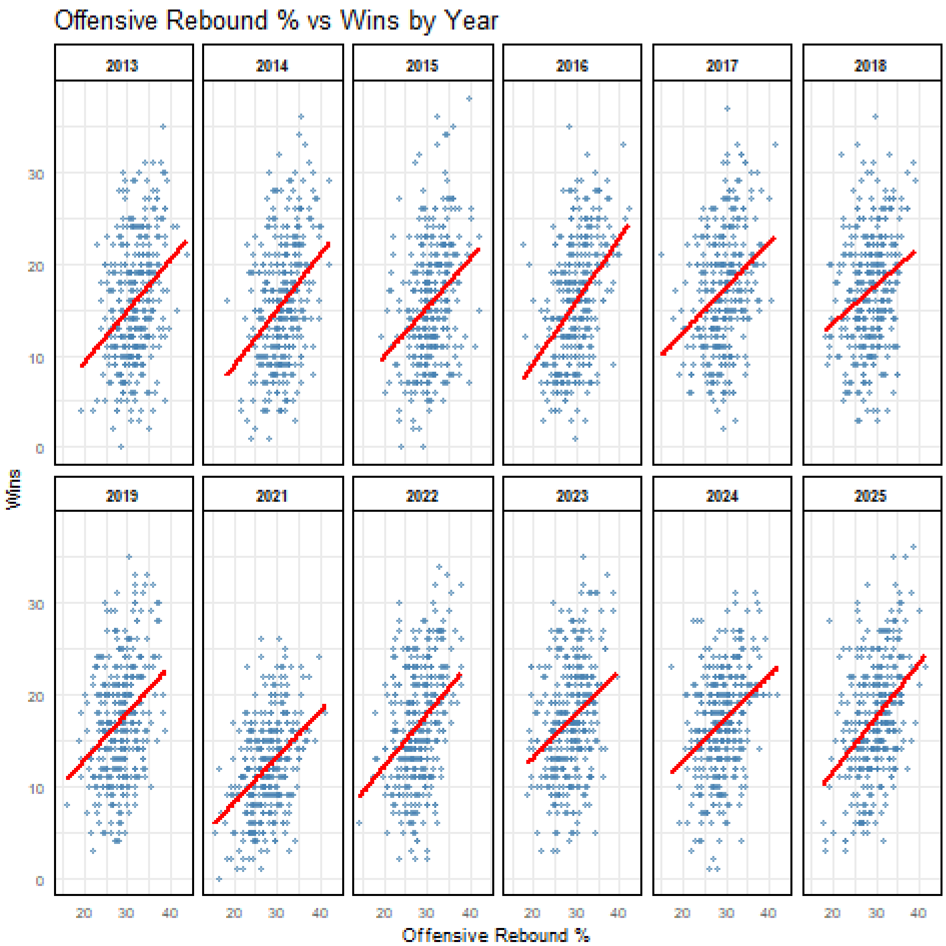
```r
# Common theme
facet_theme <- theme_minimal() +
  theme(
    strip.background = element_rect(fill = "white", color = "black"),
    strip.text = element_text(size = 8, face = "bold"),
    axis.title = element_text(size = 9),
    axis.text = element_text(size = 7),
    panel.border = element_rect(color = "black", fill = NA)
  )

# 1. Effective Field Goal % vs Wins
p1 <- ggplot(df, aes(x = EFG_O, y = W)) +
  geom_point(color = "steelblue", alpha = 0.5, size = 1) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  facet_wrap(~ YEAR, ncol = 6) +
  labs(title = "Effective FG % vs Wins by Year",
       x = "Effective FG %", y = "Wins") +
  facet_theme

# 2. Turnover % vs Wins
p2 <- ggplot(df, aes(x = TOR, y = W)) +
  geom_point(color = "steelblue", alpha = 0.5, size = 1) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  facet_wrap(~ YEAR, ncol = 6) +
  labs(title = "Turnover % vs Wins by Year",
       x = "Turnover %", y = "Wins") +
  facet_theme

# 3. Offensive Rebound % vs Wins
p3 <- ggplot(df, aes(x = ORB, y = W)) +
  geom_point(color = "steelblue", alpha = 0.5, size = 1) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  facet_wrap(~ YEAR, ncol = 6) +
  labs(title = "Offensive Rebound % vs Wins by Year",
       x = "Offensive Rebound %", y = "Wins") +
  facet_theme

# 4. Free Throw Rate vs Wins
p4 <- ggplot(df, aes(x = FTR, y = W)) +
  geom_point(color = "steelblue", alpha = 0.5, size = 1) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  facet_wrap(~ YEAR, ncol = 6) +
  labs(title = "Free Throw Rate vs Wins by Year",
       x = "Free Throw Rate", y = "Wins") +
  facet_theme

# Print all four
print(p1)
print(p2)
print(p3)
print(p4)
```

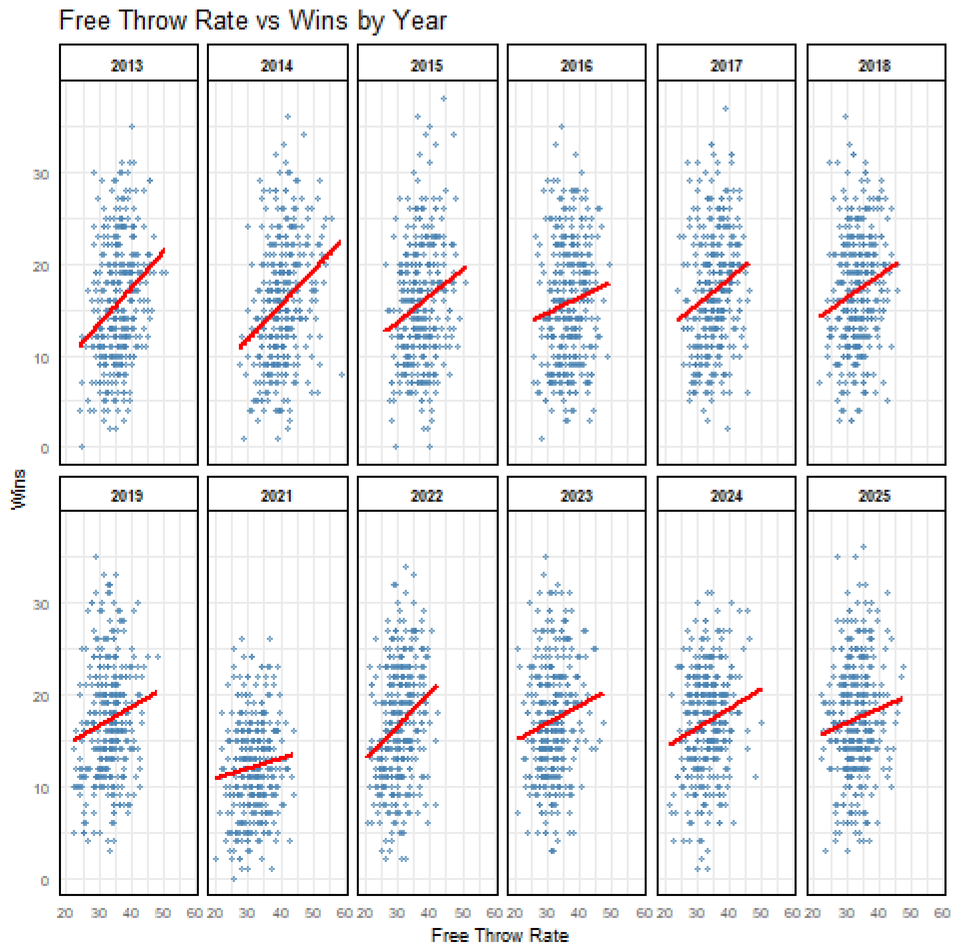
```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```





Turnover % vs Wins by Year
2013
2014
2015
2016
2017
2018
2019
2021
2022
2023
2024
2025
Wins
Turnover %
0
10
20
30
12
16
20
24


Offensive Rebound % vs Wins by Year
2013
2014
2015
2016
2017
2018
2019
2021
2022
2023
2024
2025
Wins
Offensive Rebound %
0
10
20
30
20
30
40



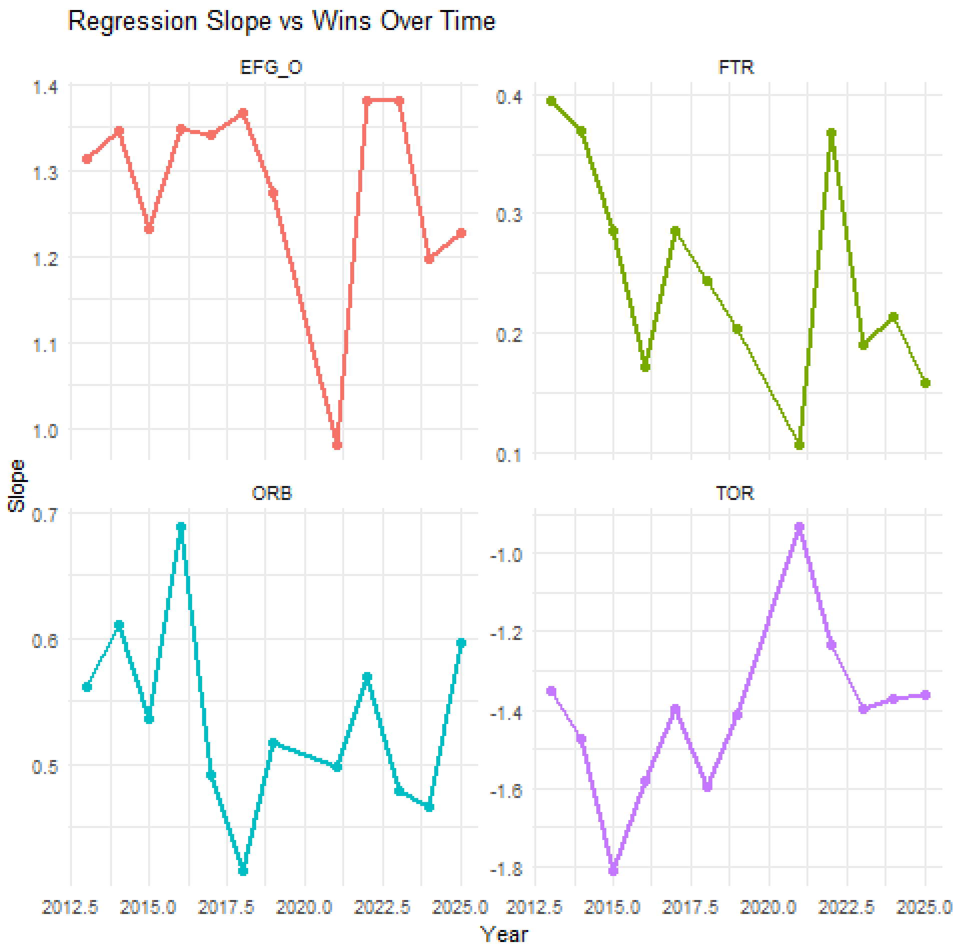
In [8]:
```
%%R -i df
library(dplyr)

vars <- c("EFG_O", "TOR", "ORB", "FTR")
years <- sort(unique(df$YEAR))

slope_results <- data.frame(Variable = character(), YEAR = numeric(), Slope = numer

for (v in vars) {
  for (y in years) {
    sub_df <- df[df$YEAR == y, ]

    # Skip if not enough data
    if (nrow(sub_df) < 3) next

    slope <- coef(lm(as.formula(paste("W ~", v)), data = sub_df))[2]

    slope_results <- rbind(slope_results, data.frame(
      Variable = v,
      YEAR     = y,
      Slope    = round(slope, 4)
    ))
  }
}

print(slope_results)
```

```
         Variable YEAR   Slope
EFG_O       EFG_O 2013  1.3130
EFG_O1      EFG_O 2014  1.3457
EFG_O2      EFG_O 2015  1.2324
EFG_O3      EFG_O 2016  1.3478
EFG_O4      EFG_O 2017  1.3400
EFG_O5      EFG_O 2018  1.3656
EFG_O6      EFG_O 2019  1.2746
EFG_O7      EFG_O 2021  0.9822
EFG_O8      EFG_O 2022  1.3808
EFG_O9      EFG_O 2023  1.3805
EFG_O10     EFG_O 2024  1.1963
EFG_O11     EFG_O 2025  1.2275
TOR           TOR 2013 -1.3524
TOR1          TOR 2014 -1.4757
TOR2          TOR 2015 -1.8096
TOR3          TOR 2016 -1.5825
TOR4          TOR 2017 -1.3957
TOR5          TOR 2018 -1.5946
TOR6          TOR 2019 -1.4106
TOR7          TOR 2021 -0.9321
TOR8          TOR 2022 -1.2313
TOR9          TOR 2023 -1.3957
TOR10         TOR 2024 -1.3701
TOR11         TOR 2025 -1.3632
ORB           ORB 2013  0.5619
ORB1          ORB 2014  0.6102
ORB2          ORB 2015  0.5366
ORB3          ORB 2016  0.6883
ORB4          ORB 2017  0.4920
ORB5          ORB 2018  0.4155
ORB6          ORB 2019  0.5175
ORB7          ORB 2021  0.4980
ORB8          ORB 2022  0.5693
ORB9          ORB 2023  0.4780
ORB10         ORB 2024  0.4655
ORB11         ORB 2025  0.5961
FTR           FTR 2013  0.3950
FTR1          FTR 2014  0.3689
FTR2          FTR 2015  0.2857
FTR3          FTR 2016  0.1723
FTR4          FTR 2017  0.2853
FTR5          FTR 2018  0.2432
FTR6          FTR 2019  0.2041
FTR7          FTR 2021  0.1070
FTR8          FTR 2022  0.3681
FTR9          FTR 2023  0.1909
FTR10         FTR 2024  0.2137
FTR11         FTR 2025  0.1582
```

In [9]:

```
%%R -i df
library(ggplot2)

ggplot(slope_results, aes(x = YEAR, y = Slope, color = Variable, group = Variable))
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
```

```
  facet_wrap(~ Variable, scales = "free_y", ncol = 2) +
  labs(title = "Regression Slope vs Wins Over Time",
       x = "Year", y = "Slope") +
  theme_minimal() +
  theme(legend.position = "none")
```



In [ ]: